

Abstract

This study investigates long-term changes in vegetation dynamics and surface temperature in Varanasi, Uttar Pradesh, using satellite-based remote sensing techniques. Multi-temporal analyses of the Normalized Difference Vegetation Index (NDVI) and Soil Adjusted Vegetation Index (SAVI) were conducted using Landsat 5 TM (1988–1994) and Landsat 8 OLI (2013–2024), applying a consistent vegetation threshold of 0.23. Seasonal Land Surface Temperature (LST) variations were derived from thermal bands, while rainfall trends were examined using CHIRPS data spanning 1980 to 2024. Additionally, Land Use and Land Cover (LULC) dynamics were assessed by comparing classified datasets from 2005–2006 and 2021, revealing significant transformations such as a rise in built-up area from 119.17 sq.km to 268.01 sq.km and a cropland expansion from 1,015.99 sq.km to over 10,811.00 sq.km. The study employed supervised classification, threshold-based filtering, and seasonal aggregation to extract robust patterns across pre-monsoon, monsoon, post-monsoon, and winter periods. Results indicate a declining vegetation trend in pre-monsoon and post-monsoon NDVI/SAVI, a noticeable increase in LST during summer seasons post-2013, and irregular rainfall patterns with both excesses and deficits during critical agricultural phases. These insights underline the intensifying anthropogenic pressure and climatic variability in the region. The integrated geospatial assessment offers crucial baselines for sustainable land management and climate adaptation strategies in urbanizing landscapes like Varanasi.

Introduction

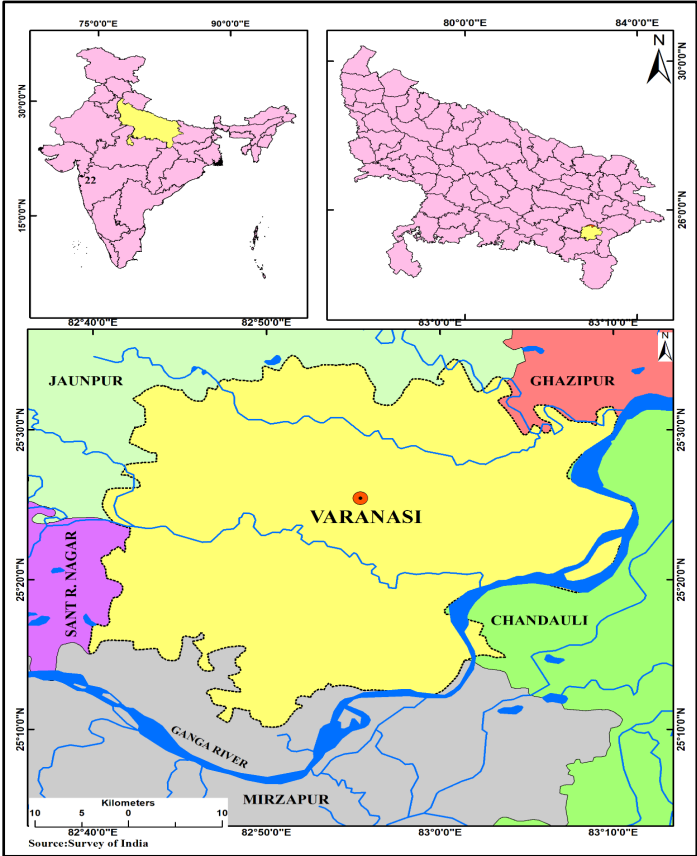
Satellite remote sensing has revolutionized the monitoring of vegetation health, land surface temperature (LST), and rainfall variability by providing consistent multi-temporal datasets across large regions (Hu et al., 2024; Roy et al., 2014). Vegetation indices such as the Normalized Difference Vegetation Index (NDVI) and the Soil-Adjusted Vegetation Index (SAVI) are widely used to assess ecological stress, agricultural productivity, and land cover change. Combined with thermal measurements, these indices offer valuable insights into the impacts of climatic variability and anthropogenic pressures, particularly in rapidly urbanizing environments (Carlson & Ripley, 1997; Piao et al., 2007).

In India, remote sensing has been extensively applied to evaluate agricultural performance, vegetation stress, and urban heat island dynamics across diverse agro-climatic zones (Dadhwal & Ray, 2000; Pathak & Rajawat, 2021). The Indo-Gangetic Plain, with its fertile soils, dense population, and accelerated urban growth, is a region where vegetation dynamics are strongly influenced by monsoonal variability and land use transitions (Sahu & Shekhar, 2015; Sannigrahi & Singh, 2019). Recent studies show that intensified urbanization in cities such as Delhi, Lucknow, and Patna has contributed to vegetation decline, rising LST, and rainfall irregularities (Saha & Bandyopadhyay, 2019; Sharma & Rao, 2021; Guha & Govil, 2020).

For Varanasi, existing research has highlighted declining vegetation indices, increasing LST, and shifting rainfall patterns (Pandey & Singh, 2022; Khanam et al., 2023). However, most analyses are limited either by short timeframes or by a narrow focus on single indicators such as NDVI or LST. Long-term, seasonally disaggregated, multi-variable studies that jointly assess vegetation, surface temperature, rainfall, and land use/land cover (LULC) change remain scarce. To address this gap, the present study integrates NDVI, SAVI, LST, CHIRPS rainfall, and multi-epoch LULC data over three decades, comparing two distinct periods—1988–1994 and 2013–2024—to provide a holistic perspective on vegetation–climate–urban interactions.

Varanasi district, located in southeastern Uttar Pradesh within the middle Indo-Gangetic Plain, covers about 1,535 km²

between 24°56′–25°35′ N and 82°14′–83°24′ E. The Ganga River forms its northern boundary, shaping geomorphology and supporting agriculture and settlement. With a population of 3.6 million and a density exceeding 2,400 persons per km² (Census of India, 2011), Varanasi has experienced rapid urban expansion, leading to vegetation loss and heightened thermal stress. The district’s humid subtropical climate is marked by hot summers (temperatures above 40°C), mild winters, and an average annual rainfall of ~1,100 mm, 85% of which falls during the monsoon season (IMD, 2020). Increasing rainfall irregularities and extreme events such as droughts and heatwaves (Rathore & Attri, 2019; Singh et al., 2022) further intensify environmental stress, making Varanasi representative case for analyzing long-term vegetation, climate, and land use dynamics in the Indo-Gangetic Plain.



Data and Methodology

This study employs multi-source geospatial datasets to analyze long-term trends in vegetation health, surface temperature, rainfall variability, and land use/land cover change across Varanasi district. All datasets were selected to ensure temporal consistency, spatial accuracy, and relevance to the study's objectives. The entire analysis is focused on satellite-derived data, primarily from the Landsat series, in conjunction with high-resolution gridded rainfall data and ancillary topographic references. The following datasets and sources were utilized:

Dataset	Input Format	Processing Steps	Output
Landsat 5 TM	Level-2 SR	Cloud masking, SR conversion, NDVI/SAVI, LST	Seasonal maps(1988–1994)
Landsat 8/9 OLI/TIRS	Level-2 SR	Same as above	Seasonal maps(2013–2024)
CHIRPS Rainfall	Daily TIFF	Aggregation, anomalies, SPI	Time series, maps (1980–2024)
All Datasets	Geo-TIFF	CRS unification, clipping, resampling	Aligned layers for analysis

Table 1datasets and sources

To perform a consistent temporal analysis, two distinct time periods were selected:

- **Epoch I (1988–1994) :** Landsat 5 Thematic Mapper (TM) imagery
- **Epoch II (2013–2024):** Landsat 8 Operational 2Land Imager (OLI) and Thermal Infrared Sensor (TIRS)

All scenes were sourced via the USGS Earth Explorer and processed on Google Earth Engine (GEE). Only cloud-free and seasonally comparable images— season months (November primarily from dry- to March)—were selected to minimize the effects of phenological variation and monsoon-induced greenness spikes. Image selection followed standard pre-processing practices, including radiometric calibration, cloud masking (QA_PIXEL band), and conversion to surface reflectance where applicable.

This study adopts a comprehensive geospatial and statistical approach to assess multi-temporal dynamics of vegetation health, land surface temperature (LST), rainfall variability, and land use/land cover (LULC) transformation in the Varanasi region. It integrates remotely sensed datasets from Landsat missions, CHIRPS precipitation records, and supervised LULC classifications. The methodology is structured into five primary components: vegetation index analysis, LST analysis, rainfall variability assessment, LULC classification and change detection, and correlation-based integration. Each component is described below in detail.

3.1 Vegetation Index Analysis

To analyse long-term vegetation health, two vegetation indices were employed: the Normalized Difference Vegetation Index (NDVI) and the Soil-Adjusted Vegetation Index (SAVI). NDVI is computed as:

NDVI = (NIR - Red) / (NIR + Red)

where NIR and Red represent surface reflectance values in the near-infrared and red bands respectively. To account for soil background effects in sparsely vegetated areas, SAVI was additionally used, defined as:

SAVI = ((NIR - Red) / (NIR + Red + L)) × (1 + L)

where L is a canopy background adjustment factor, set to 0.5 as recommended for intermediate vegetation densities (Huete, 1988).

The NDVI and SAVI indices were derived from surface reflectance-corrected Landsat 5 TM (1988–1994) and Landsat 8 OLI (2013–2024) imagery. Cloud and shadow masking were applied using the pixel quality assurance bands. Only pixels with NDVI and SAVI values exceeding the minimum threshold of 0.23 were retained for analysis to ensure biologically relevant vegetation signals. For each period (1988–1994 and 2013–2024), statistical summaries of NDVI and SAVI were computed, including the mean, median, and standard deviation.

These statistics offer insights into central tendencies and variability across both time spans. Tables presenting these summary statistics were generated to compare spatially aggregated vegetation conditions across the two periods. To assess spatial dynamics, difference maps were generated for both indices by subtracting the earlier period from the later one (e.g., ΔNDVI = NDVI_2013–2024–NDVI_1988–1994).These maps highlight zones of vegetation degradation or improvement, enabling the identification of ecological hotspots undergoing significant change. Given the monsoonal climate and variability in image acquisition dates, seasonal composites were created for consistent comparison between the two periods. Four standard seasons were defined based on the regional climatic cycle which is Pre-monsoon (March–May), Monsoon (June–September),Post-monsoon (October–November), and Winter (December–February).

Further, a time series analysis was conducted using all available years within each period. The Sen’s slope estimator was employed to determine the magnitude of temporal trends in NDVI and SAVI, while the Mann-Kendall test (non-parametric) was used to evaluate their statistical significance. Both methods are robust against non-normal data distributions and missing values. These trend metrics were computed for seasonal and annual composites to distinguish short-term phenological variations from long-term changes.

3.2 Land Surface Temperature (LST) Analysis

Land Surface Temperature (LST) was retrieved from the thermal infrared bands of Landsat 5 TM (Band 6) and Landsat 8 OLI/TIRS (Band 10) using the Radiative Transfer Model (RTM), a well-established method for satellite-based surface temperature estimation (Sobrino et al., 2004; Jiménez-Muñoz & Sobrino, 2003). The preprocessing workflow involved converting the digital number (DN) values into Top-of-Atmosphere (TOA) spectral radiance using radiometric rescaling coefficients provided in the metadata. Subsequently, at-sensor brightness temperature (T_a) was derived using the thermal calibration constants K₁ and K₂ specific to each sensor, following the Planck’s inverse function:

T_{sensor} = K₂ / ln (L_A / K₁ + 1)

where Lλ denotes TOA spectral radiance (W/m²·sr·μm), and K₁ and K₂ are thermal calibration constants obtained from Landsat metadata.

To account for surface emissivity (ϵ), a critical factor in accurately estimating LST, the NDVI-based emissivity correction method was applied, as recommended by [Sobrino et al. \(2004\)](#). The proportion of vegetation (P_v) was calculated from NDVI values, and ϵ was then estimated using:

$$\epsilon = \epsilon_v \times P_v + \epsilon_s \times (1 - P_v) + C$$

Here, ϵ_v and ϵ_s represent emissivity values for vegetation (typically ~ 0.99) and bare soil (~ 0.96), respectively, and C is a surface roughness correction term. The proportion of vegetation (P_v) was computed as:

$$\text{where } P_v = \left(\frac{\text{NDVI} - \text{NDVI}_{\min}}{\text{NDVI}_{\max} - \text{NDVI}_{\min}} \right)^2$$

with $\text{NDVI}_{\min}$ and $\text{NDVI}_{\max}$ determined empirically (usually set to 0.2 and 0.5, respectively, as per [Sobrino et al., 2004](#)).

LST was then calculated using the corrected emissivity and brightness temperature via the generalized Planck's function adapted for surface temperature:

$$LST = \frac{T_{\text{sensor}}}{1 + \rho \cdot \lambda \cdot \left(\frac{T_{\text{sensor}}}{K} \right) \cdot \ln(\epsilon)}$$

where λ is the wavelength of emitted radiance (centered at 11.5 μm for Band 10), and ρ is a constant derived from Planck's equation ($\rho = h \cdot c / \sigma = 1.438 \times 10^{-2} \text{ m} \cdot \text{K}$). Final LST values were expressed in degrees Celsius for interpretability.

For each epoch (1988–1994 and 2013–2024), seasonal mean daytime LST maps were generated using cloud-free composite images. These maps facilitated both spatial and temporal analysis. Difference images (ΔLST) were created to highlight spatial shifts in surface thermal regimes between the two periods. Regions consistently exhibiting LST increases were interpreted as potential urban heat island (UHI) zones, following methods outlined by [Weng et al. \(2004\)](#).

To quantify temporal trends in surface warming, non-parametric trend analysis techniques—Sen's slope estimator and the Mann-Kendall test—were applied to the seasonal LST datasets, similar to the approach used for NDVI and SAVI analyses ([Zhou et al., 2016](#)). To further investigate the UHI effect, spatial zonation was performed using land use/land cover (LULC) classifications to delineate densely built-up versus vegetated zones. Mean seasonal LST values were then compared across these land cover categories to assess thermal disparities attributable to urban expansion and vegetation loss.

The resulting seasonal LST maps and associated metrics served as critical inputs for integrated analyses of vegetation–temperature dynamics, including NDVI–LST correlation assessments and spatiotemporal UHI detection across the multi-decadal interval.

3.3 Rainfall Variability Analysis

Rainfall data for the study period (1980–2024) were obtained from the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) product, which provides quasi-global coverage at a spatial resolution of approximately 0.05° ($\sim 5.5 \text{ km}$) and daily temporal granularity ([Funk et al., 2015](#)). For the Varanasi study region, daily rainfall values were aggregated to monthly, seasonal, and annual totals using zonal statistics. The seasonal classification was consistent with other variables in the

study, defined as follows: winter (December–February), pre-monsoon (March–May), monsoon (June–September), and post-monsoon (October–November).

To evaluate temporal rainfall dynamics and interannual variability, time series were constructed for annual and seasonal totals. Linear regression and non-parametric trend estimation using Sen's slope were applied to quantify the direction and magnitude of rainfall trends over the study period. Additionally, the Mann-Kendall test was employed to assess statistical significance of these trends without assuming normality of the underlying data distribution ([Yue & Wang, 2002](#)).

Rainfall anomalies were computed to detect deviations from climatological normal. The standardized rainfall anomaly for a given year t was calculated as:

$$\text{Anomaly} = P_t - \bar{P}$$

Where P_t is rainfall at time t , and $\bar{P}$ is the multi-decadal mean.

Years with anomalies exceeding ± 1 standard deviation were identified as hydrological extremes: values below -1 indicating drought years, and values above $+1$ suggesting excessive rainfall or flood conditions. These anomalies were spatially mapped and temporally aligned with vegetation indicators (NDVI and SAVI) to examine potential eco-climatic interactions.

Furthermore, the Standardized Precipitation Index (SPI) was computed to characterize precipitation deficits and surpluses on an annual scale.

The SPI for a given year t is defined as:

$$SPI = \frac{P_t - \bar{P}}{\sigma}$$

Where σ is the standard deviation of long-term rainfall. This formulation is statistically equivalent to the standardized anomaly, but its use aligns with established drought monitoring protocols ([McKee et al., 1993](#)).

To detect potential shifts in seasonal rainfall patterns—especially related to the timing and intensity of the Indian monsoon—the Seasonal Mann-Kendall test was applied separately to each seasonal subset (e.g., monsoon-only series), facilitating identification of intra-annual trend behaviour. Together, these methods provided a comprehensive assessment of precipitation variability, long-term trends, and extremes, enabling detailed investigation of their implications for vegetation health and surface temperature dynamics in the Varanasi region.

3.4 Land Use/Land Cover (LULC) Classification and Change Detection

Land use and land cover (LULC) transformation was assessed through supervised classification of medium-resolution satellite imagery acquired for four representative time points: 1994, 2004, 2014, and 2024. To maximize surface visibility and reduce atmospheric interference, images were selected from the dry season, when cloud cover is minimal and spectral separability among land cover classes is enhanced.

Two algorithms—Maximum Likelihood Classifier (MLC) and Random Forest (RF)—were tested. The RF classifier was ultimately adopted owing to its higher overall accuracy, robustness against overfitting, and proven performance in LULC

mapping (Belgiu & Drăguț, 2016). Training samples for urban, cropland, forest, water, barren land, and other categories were generated through visual interpretation of high-resolution reference imagery, with 50–100 ground-truth points per class to ensure stability and representativeness.

Classification results were validated using an independent set of ground reference points. A confusion matrix was constructed to evaluate performance, and user’s accuracy, producer’s accuracy, overall accuracy, and the Kappa coefficient were computed. All metrics indicated satisfactory reliability, with classification accuracy exceeding 85%.

To quantify temporal change, area statistics were extracted for each LULC class. Change detection was conducted through transition matrices, capturing inter-class conversions such as cropland-to-urban expansion and wetland-to-barren shifts. These changes were further illustrated with color-coded LULC maps to support spatial interpretation of transformation patterns.

3.5 Correlation, Trend Detection, and Integration of Variables

Statistical analysis was undertaken to integrate vegetation indices, climate, and thermal indicators. NDVI and LST were compared across representative years to examine their expected inverse relationship in urbanized zones. Scatterplots and correlation coefficients (Pearson’s *r* and Spearman’s ρ) were calculated to quantify the strength and direction of associations.

In addition, NDVI and SAVI were correlated with seasonal rainfall anomalies to assess vegetation responses to climatic variability. Both annual and seasonal relationships were examined, building on earlier studies that used NDVI as a proxy for vegetation productivity under rainfall stress (Pettorelli, 2013; Pandey & Singh, 2022).

The following statistical approaches were applied:

Pearson’s correlation → to assess linear relationships

Spearman’s rank correlation → to assess monotonic but non-linear relationships

Mann–Kendall trend test → to detect the statistical significance of monotonic trends in NDVI, SAVI, LST, and rainfall time series (Kendall, 1975)

Sen’s slope estimator → to quantify the magnitude of trends (Sen, 1968)

Significance thresholds were set at $p < 0.05$, and confidence intervals were reported where applicable. By jointly analyzing vegetation indices, LST, rainfall, and LULC, this framework provides a multi-dimensional understanding of environmental change in Varanasi, enhancing robustness through the integration of remote sensing and statistical methods.

Results

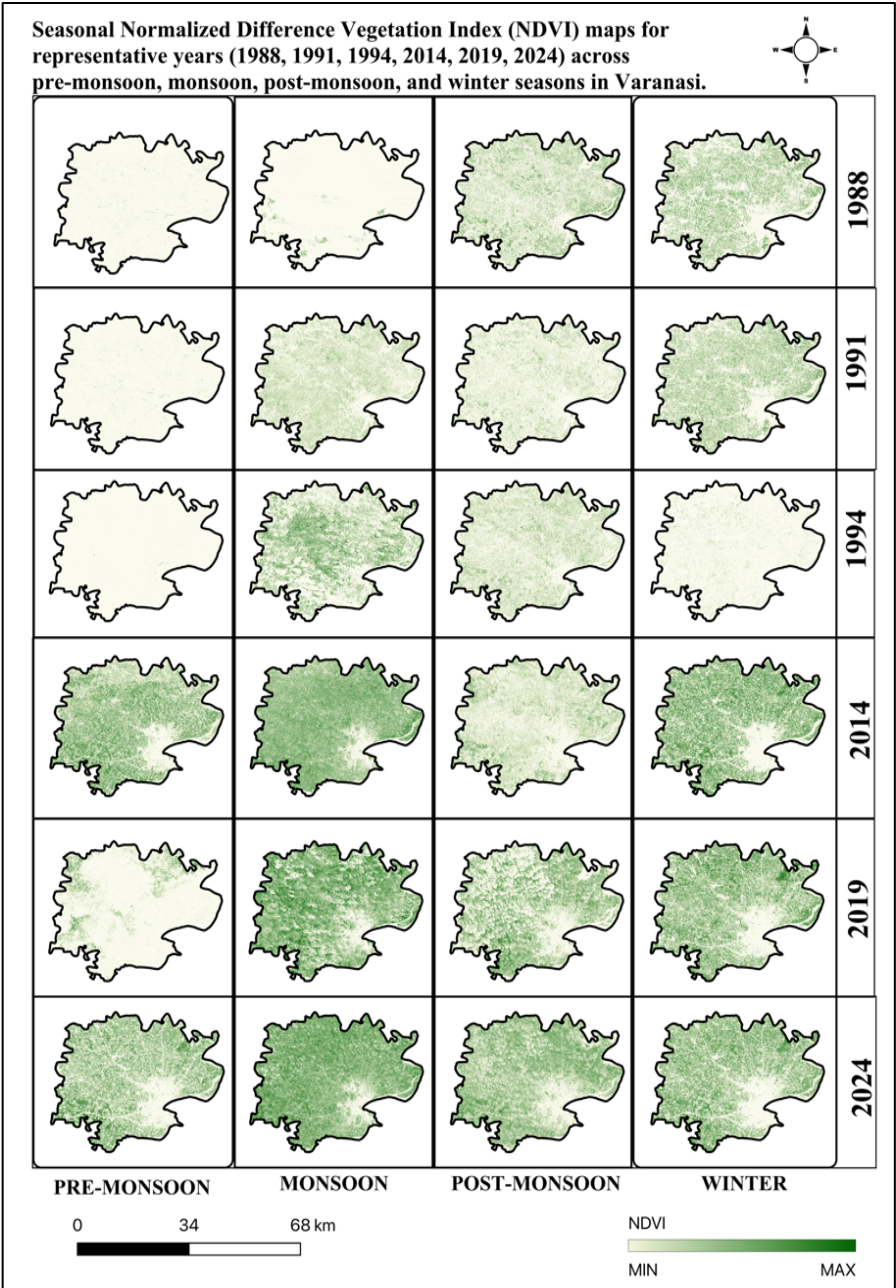
This section presents the results of the multi-decadal analysis of vegetation health (NDVI and SAVI), land surface temperature

(LST), rainfall variability, and land use/land cover (LULC) changes in Varanasi district. The datasets span from 1988 to 2024 for vegetation and LST, and from 1980 to 2024 for rainfall, while LULC classification maps are presented for 1994, 2004, 2014 and 2024. All outputs are derived from pre-processed satellite imagery, following rigorous thresholding, spatial harmonization, and temporal aggregation procedures.

4.1 Vegetation Indices (NDVI and SAVI)

4.1.1 Spatial Patterns and Temporal Trends

The spatial distribution of mean NDVI and SAVI during the two



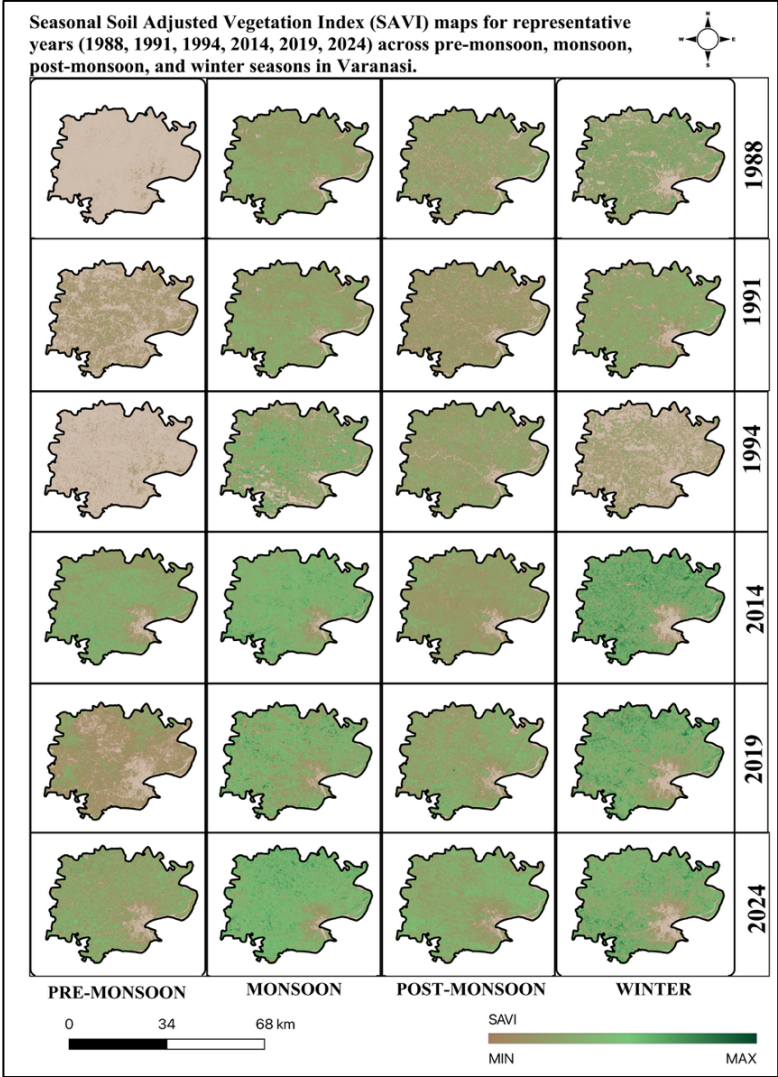
study windows—1988–1994 and 2014–2024—reveals distinct seasonal and spatial contrasts.

Higher values are consistently observed along riparian belts, wetlands, and cultivated areas, while urban cores and barren zones display low index values. A noticeable decline in both NDVI and SAVI is evident in expanding built-up regions, suggesting the progressive replacement of vegetation by impervious surfaces.

Statistic	NDVI (1988–1994)	NDVI (2013–2024)	SAVI (1988–1994)	SAVI (2013–2024)
Mean	0.41	0.34	0.46	0.38
Max	0.78	0.74	0.82	0.77
Std Dev	0.14	0.11	0.15	0.13

Table 2-Spatial Patterns

The annual time series of NDVI and SAVI (1988–2024) exhibits a gradual but statistically significant decline, most pronounced after 2010 when urban expansion accelerated. The Mann–



Kendall test confirms a negative trend ($p < 0.05$), while Sen’s slope indicates annual reductions of -0.004 (NDVI) and -0.005 (SAVI). Seasonal analysis shows the sharpest declines during pre-monsoon and post-monsoon periods, coinciding with peak water stress and intensified land conversion.

4.2 Land Surface Temperature (LST)

4.2.1 Spatial Distribution and Trends

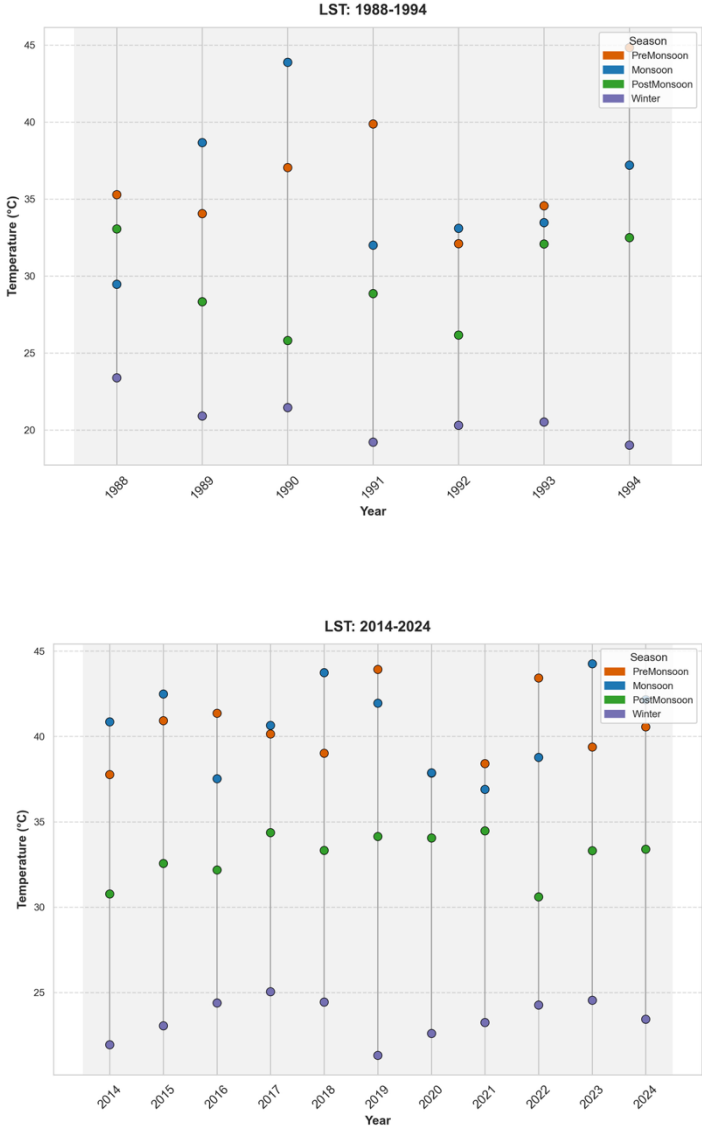
Mean daytime LST for 1988–1994 and 2013–2024 demonstrates a clear rise in surface temperatures, particularly in central urban areas. The LST difference map (Δ LST) indicates localized increases of 4–5°C, consistent with expansion of built-up surfaces and loss of vegetative cover.

Period	Mean LST (°C)	Max LST (°C)	Std Dev (°C)
1988–1994	28.7	39.3	3.2

2013–2024	32.4	42.7	2.9
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Table 3. Summary of mean daytime LST values for the two epochs.

The long-term time series (1988–2024) reveals a persistent warming trajectory. Sen’s slope indicates an annual increase of approximately $+0.15^{\circ}\text{C}$, while the Mann–Kendall test confirms a statistically significant positive trend ($p < 0.01$).



4.2.2 Urban Heat Island (UHI) Assessment

Period	Built-up (°C)	Vegetation (°C)	UHI Intensity ($\Delta^{\circ}\text{C}$)
1988–1994	30.9	28.7	+2.2
2013–2024	34.7	30.9	+3.8

Table 4-Comparison of LST

Comparison of LST in built-up versus vegetated zones highlights a strengthening UHI effect. In 1988–1994, built-up areas were on average 2.2°C warmer than vegetated zones; this gap widened to 3.8°C in 2013–2024. The intensification reflects the cumulative effects of impervious surface expansion and vegetation loss within the urban core.

4.3 Rainfall Variability

YEAR	MONSOON	POST-MONSOON	PRE-MONSOON	WINTER
1981	954.9229	18.1366	30.7405	67.3644
1982	790.0713	17.4627	47.9679	23.7048
1983	770.6586	49.3265	39.7497	44.8149
1984	841.4581	22.4397	30.7622	28.7358
1985	1168.2722	99.7295	22.4643	40.038
1986	982.6245	21.5487	28.1502	27.4273
1987	1010.6402	44.0232	31.258	25.6043
1988	1011.8857	28.4836	37.7657	27.0386
1989	962.3839	22.0153	29.2669	42.0266
1990	1105.5322	21.3322	50.3636	25.1059
1991	894.0683	12.0224	32.5831	24.4634
1992	464.4743	41.1065	30.1798	19.8893
1993	658.7128	29.2648	43.0004	36.5565
1994	1008.6523	20.9459	23.1726	35.1933
1995	752.113	14.5043	36.6954	48.6835
1996	936.6767	51.5017	18.7281	21.5184
1997	804.8825	31.3131	38.4855	26.3232
1998	863.3477	41.9181	43.0496	31.5872
1999	1031.3758	55.7356	43.8805	22.4608
2000	957.4487	11.2515	27.981	17.1838
2001	851.3214	88.5322	31.533	32.4826
2002	637.8426	35.8943	27.5571	39.1712
2003	889.7213	21.8161	20.9449	32.4972
2004	745.5144	22.665	24.5658	45.2539
2005	681.3299	24.0537	30.5902	15.4439
2006	795.5563	26.7235	53.213	56.469
2007	849.8666	22.2492	49.4844	24.7932
2008	1097.992	12.4048	33.3761	16.3311
2009	535.3235	34.4688	31.9837	25.5395
2010	696.7536	40.6596	23.3201	18.5401
2011	975.3693	13.3738	31.6378	35.5475
2012	966.9598	27.1092	22.3506	48.0478
2013	925.2137	119.0603	30.4849	64.0856
2014	693.5051	61.8211	31.6377	29.0456
2015	613.5782	19.1822	40.0388	16.6345
2016	1195.2035	32.6429	50.1251	16.977
2017	920.0237	14.7048	25.3919	19.7179
2018	822.4725	16.4124	29.1151	34.8192
2019	1057.4562	52.7618	27.502	47.7218
2020	1090.0998	28.7602	96.4599	15.6654
2021	1060.2725	72.9921	71.4103	35.79
2022	813.7752	78.7157	18.3674	14.4606

2023	629.2565	67.0982	66.2808	41.5228
2024	992.2113	23.3721	27.936	14.924

Table 5- PRECIPITATION

4.3.1 Long-Term Trends

CHIRPS rainfall data (1980–2024) indicate substantial interannual variability, with total annual precipitation averaging 1,120 mm. A weak positive trend is suggested by Sen’s slope (+1.2 mm/year), but this is not statistically significant ($p > 0.05$). Monsoonal rainfall remains dominant, contributing more than 80% of the annual total.

4.3.2 Seasonal Breakdown and Extremes

Seasonal analysis highlights variability outside the monsoon season. Winter and post-monsoon rainfall exhibit stronger fluctuations compared to monsoon totals. Several drought years—1987, 1992, 2002, and 2015—recorded rainfall deficits exceeding 25% of the long-term mean, while flood-prone years such as 2013 and 2021 correspond to excessive monsoonal precipitation. These extremes likely influenced vegetation cycles and surface thermal dynamics in the district.

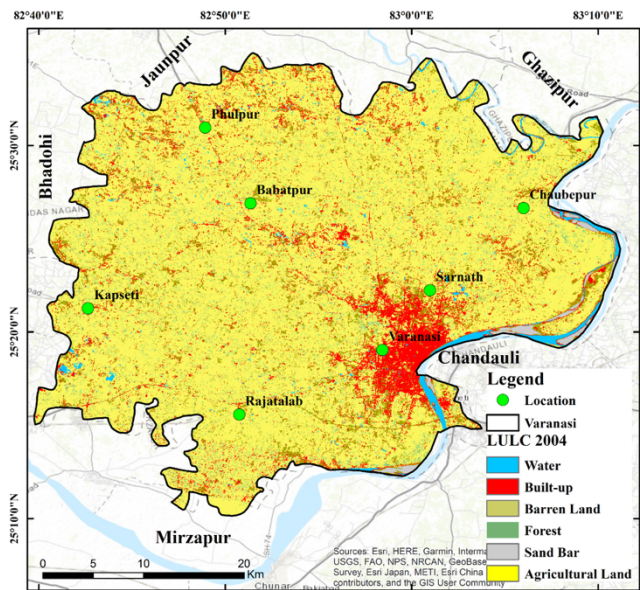
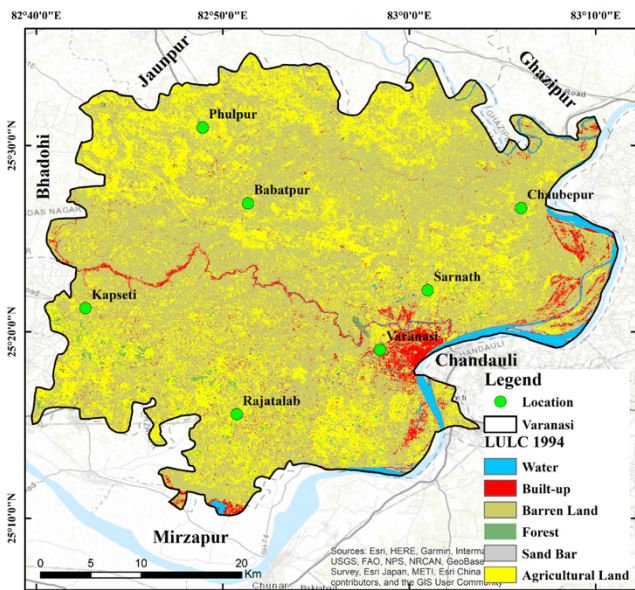
4.3.3 Trend Analysis (Mann–Kendall and Sen’s Slope)

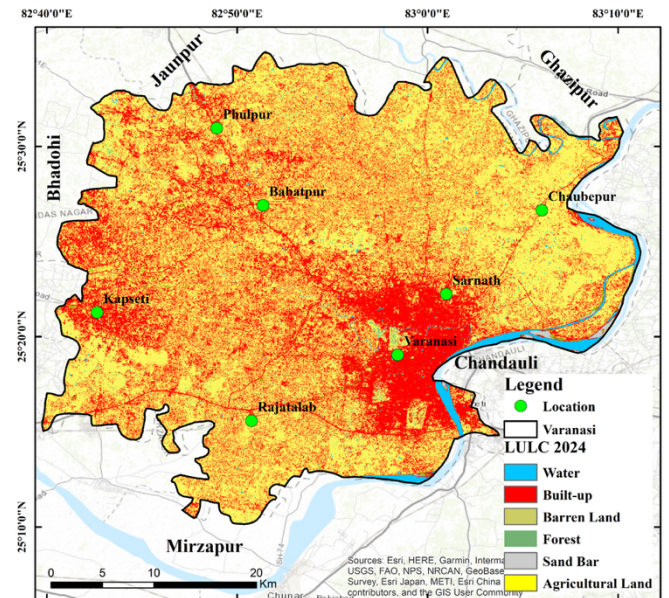
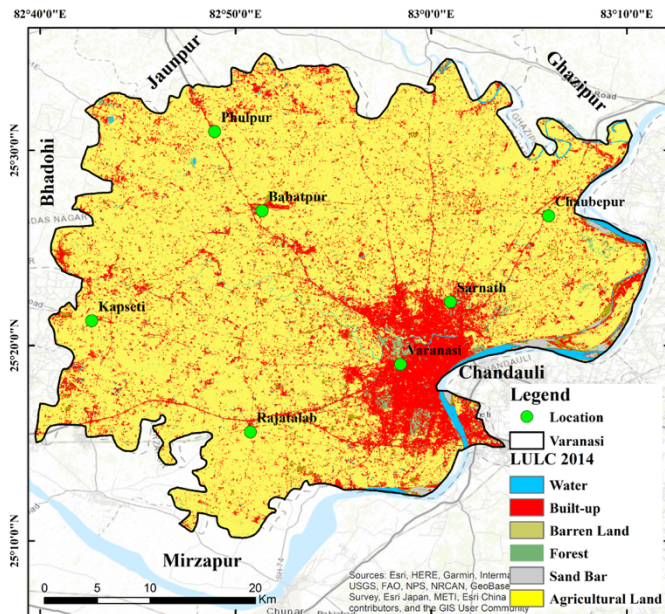
To quantify long-term trends, the non-parametric Mann–Kendall test and Sen’s slope estimator were applied to the NDVI, SAVI, LST, and rainfall series. Results indicate statistically significant negative vegetation trends, a significant positive warming trend, and a non-significant upward trend in rainfall

Variable	Period	Sen’s slope	Z-statistic	p-value	Trend direction
NDVI	1988–2024	−0.004	−2.1	0.035	Negative
SAVI	1988–2024	−0.005	−2.4	0.018	Negative
LST	1988–2024	+0.150	+3.2	0.001	Positive
Rainfall	1980–2024	+1.200	+1.1	0.270	Non-significant

Table 6- Trend Analysis (Mann–Kendall and Sen’s Slope)

7.4 Land Use and Land Cover (LULC) Dynamics





4.4.1 Classification and Accuracy

LULC maps for 1994, 2004, 2014, and 2024 were generated using Random Forest classification. Training samples were derived from high-resolution imagery, with independent ground-truth data used for validation. The 2024 classification achieved an overall accuracy of 88.2% and a Kappa coefficient of 0.85, confirming reliability for subsequent change analysis.

Class	User Accuracy (%)	Producer Accuracy (%)
Built-up	91.5	89.2
Cropland	87.3	90.1
Forest/Tree cover	85.6	83.4
Water bodies	94.8	92.1
Others	82.7	80.3

Table 7-Accuracy assessment for 2021 LULC classification.

d Kappa coefficient of 0.85.

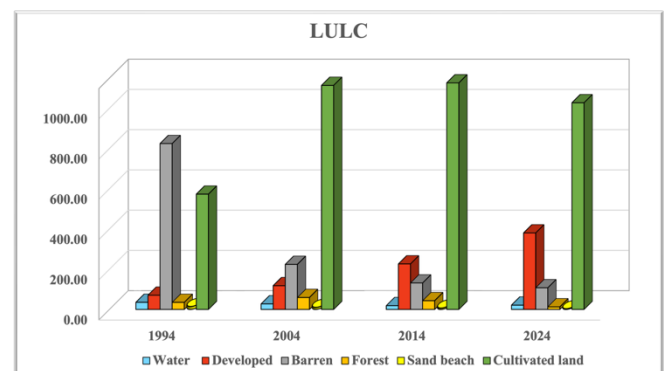
4.4.2 Area Changes and Transition Matrix

LULC statistics demonstrate substantial transformations in Varanasi over the three-decade period. Built-up land expanded more than fivefold, increasing from 70.6 km² in 1994 to 380.1 km² in 2024, primarily through the conversion of cropland and barren land. Cultivated land exhibited a two-phase trend—growing between 1994 and 2014 but declining thereafter—while forest cover experienced a sharp reduction, shrinking to only 12.8 km² in 2024.

The transition matrix highlights the most significant conversions: nearly one-third of scrub and fallow areas and one-quarter of cropland shifted into built-up surfaces, underscoring the intensity of urbanization. Cropland nonetheless retained dominance, with 63.2% of its area remaining stable, while forest cover showed greater persistence (76.3%) but at a much smaller absolute scale. Overall, the results reveal a landscape increasingly reshaped by urban growth, with notable trade-offs in agricultural and ecological land categories.

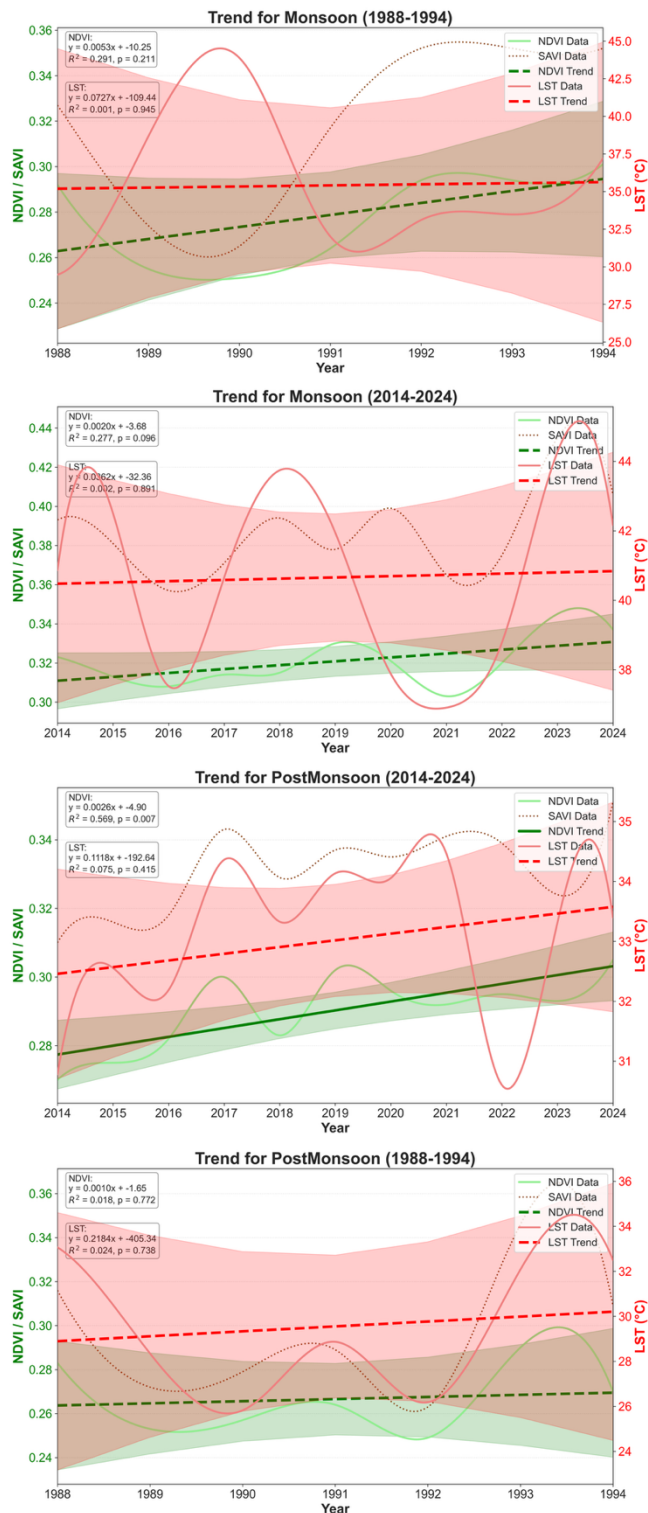
From → To	Built-up	Cropland	Forest	Water	Other
Cropland	24.8%	63.2%	2.1%	1.3%	8.6%
Forest	7.2%	9.8%	76.3%	2.3%	4.4%
Scrub/Fallow	31.5%	28.4%	6.5%	1.7%	31.9%

Table 8 - Sample LULC transition matrix .



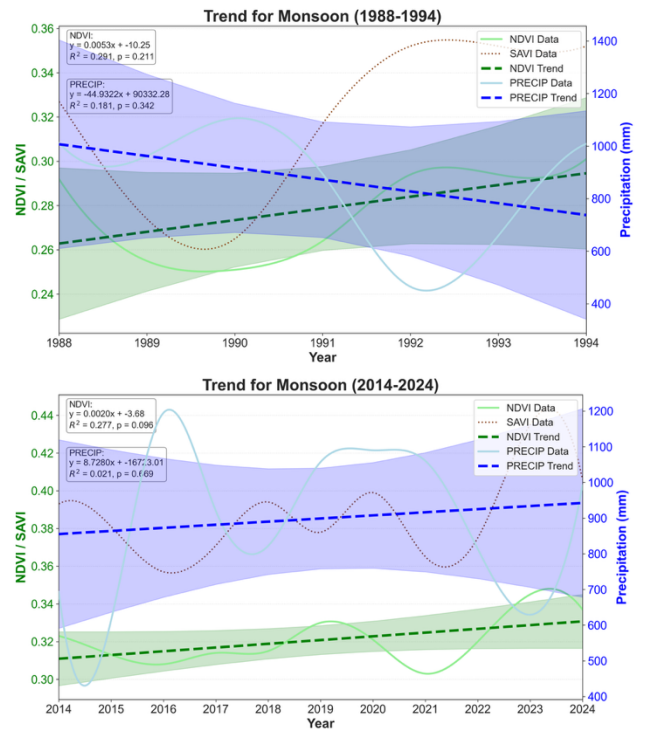
4.5 Cross-Variable Integration

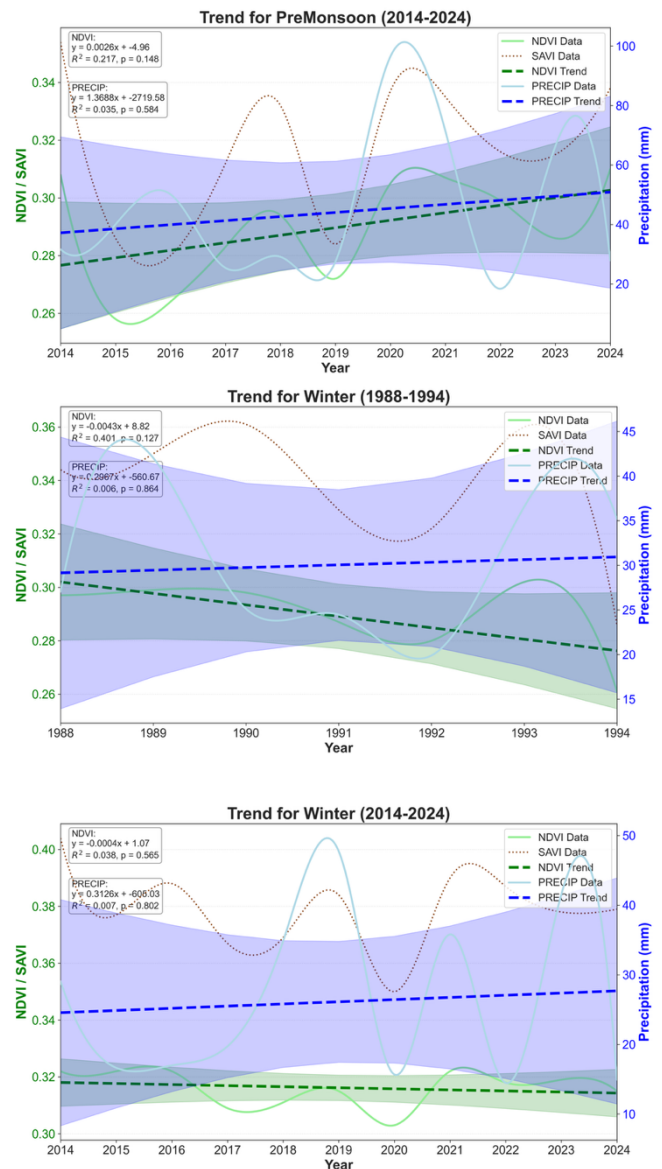
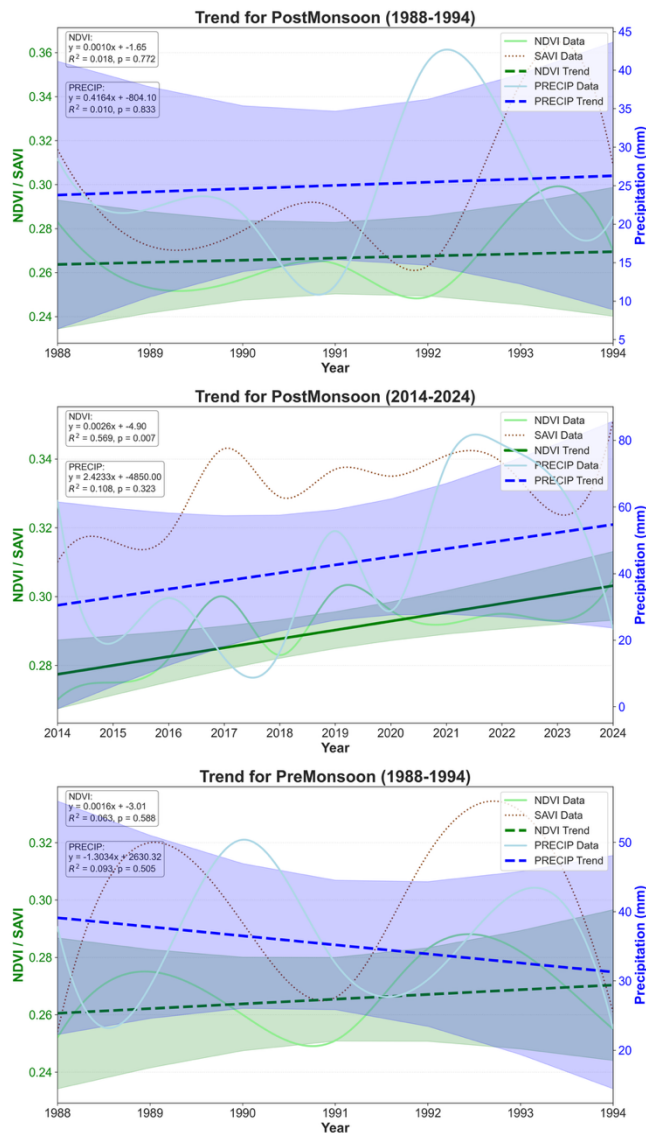
4.5.1 NDVI-LST Relationship



4.5.2 NDVI-Rainfall Linkages

Correlation analysis between NDVI and annual rainfall for overlapping years indicates a moderately positive relationship, with a Spearman's rank coefficient of 0.54 ($p < 0.05$). NDVI anomalies plotted against rainfall anomalies further show synchrony in vegetation response to moisture availability, particularly during monsoon seasons.





Key spatial observations include:

- Vegetation declines areas often overlap with zones of LULC transition (e.g., cropland to urban).
- Urban expansion corridors exhibit reduced NDVI and elevated LST values.
- Periods of low rainfall correspond to suppressed NDVI values, indicating drought sensitivity in vegetation.

Discussion

8.1 Vegetation Dynamics: NDVI and SAVI Insights

The spatiotemporal patterns of vegetation in Varanasi district over the study period (1988–2024) reveal notable changes, as captured by NDVI and SAVI indices. The average NDVI and SAVI values for the period 1988–1994 were significantly higher compared to 2013–2024, indicating a general decline in vegetative health and coverage. This decline is particularly evident in the central and eastern zones of the study area, where urban expansion has been most prominent. The line graph of annual NDVI from 1988 to 2024 illustrates a gradual but steady downward trend, with noticeable dips in vegetation indices after 2010, coinciding with the onset of rapid urban development and infrastructure expansion.

Both NDVI and SAVI metrics highlight similar trends, although SAVI provided slightly better differentiation in sparsely vegetated areas due to its soil brightness correction component. The use of a threshold value of 0.23 ensured that only significant vegetation cover was analysed, filtering out bare soil and urban features that might have confounded interpretation.

Vegetation loss is especially prominent in regions previously classified as cropland or fallow land. While there is some evidence of plantation or horticultural expansion on the peripheries (marginal increases in NDVI values in selected rural belts), these gains are marginal and insufficient to counterbalance the widespread degradation occurring due to impervious surface growth. This degradation is consistent with the findings of [Khanam et al. \(2023\)](#), who also reported declining NDVI in Varanasi between 2000 and 2020 due to urban encroachment and reduction in cropland extent [[ijisrt.com](#)].

These trends highlight the increasing vulnerability of vegetative systems to anthropogenic pressures. The loss of vegetation not only reduces ecosystem services such as carbon sequestration and soil stabilization but also has critical implications for regional microclimate regulation and biodiversity.

8.2 Thermal Environment Shifts: LST Trends and Urban Heat Island Effect

Land Surface Temperature (LST) results further reinforce the narrative of ecological stress. Between 1988–1994 and 2013–2024, there was a clear increase in average LST across the district. The average temperature rose by several degrees Celsius in key built-up areas, with urban “hotspots” such as Varanasi city centre showing the most pronounced warming. Seasonal LST analysis showed that summer and pre-monsoon temperatures have particularly intensified over time, potentially exacerbating heat stress and impacting public health.

The inverse relationship between NDVI and LST is particularly telling. Areas with higher vegetation coverage exhibited lower surface temperatures due to evapotranspiration and shading effects, while built-up and barren lands showed elevated temperatures. The scatter plot of NDVI vs. LST for a representative year (e.g., 2018) confirms this negative correlation, underscoring the role of vegetation in mitigating urban heat island (UHI) effects.

Urban expansion, as evident from LULC maps, has contributed to this thermal stress. Increased impervious surfaces (concrete, asphalt, rooftops) have replaced vegetative cover, reduced latent heat flux and enhanced sensible heat emissions. These changes are in alignment with studies from other urban regions in India, such as Lucknow and Delhi, where rapid urban growth has been closely linked to UHI intensification (Sahana et al., 2020; Roy et al., 2021).

This rise in LST poses challenges for sustainable development, especially considering the increased frequency of heatwaves in North India. Urban planning must therefore integrate green infrastructure and passive cooling strategies to combat the warming trend.

8.3 Rainfall Variability and Its Impact

CHIRPS rainfall data from 1980 to 2024 indicates considerable interannual variability, with an overall slight declining trend in monsoonal precipitation, especially post-2000. The fitted trend line in the annual rainfall time-series chart suggests a marginal negative slope, implying reduced rainfall reliability over time. Seasonal breakdown charts confirm that monsoon rainfall has become slightly less intense, whereas winter rainfall remains relatively stable.

This reduction in precipitation, though not dramatic, has implications for vegetation dynamics. Lower rainfall during critical growth periods stresses crops and natural vegetation, potentially contributing to reduced NDVI and SAVI values in affected years. For instance, significant rainfall dips in 2002, 2009, and 2015 coincide with observed NDVI troughs.

Literature supports this correlation: [studies in MDPI \(2022\)](#) emphasize that NDVI often shows a strong dependency on rainfall, particularly in semi-arid and monsoonal regions [mdpi.com]. A reduction in monsoon rainfall can limit biomass growth, delay sowing periods, and increase the risk of crop failure. Given that Varanasi is largely agrarian in its outer zones, rainfall variability directly affects livelihoods and food security.

Although the long-term trend in rainfall may not be statistically significant, the increasing irregularity and unpredictability of rainfall events—short bursts of intense rain interspersed with prolonged dry spells—pose serious challenges. This necessitates resilient agricultural practices and water resource

management strategies to buffer the impacts of hydrological uncertainty.

8.4 Land Use and Land Cover Transitions

LULC analysis between 2005–06 and 2021 reveals substantial landscape transformation. Classified maps show a marked decline in agricultural lands—particularly cropland—and a parallel expansion of built-up areas. The bar chart comparing LULC class areas quantifies this trend: cropland reduced by over 200 sq.km, while built-up land increased from 119.17 sq.km (urban + rural) to 268.01 sq.km by 2021.

The LULC transition matrix further illustrates these changes. A significant portion of cropland (~10%) transitioned into built-up land, and some scrublands and wetlands were also absorbed into urban infrastructure. Water bodies and wetlands showed minor shrinkage, likely due to encroachment or sedimentation, while tree cover slightly expanded, potentially due to plantation drives or changes in classification accuracy.

These transitions reflect both developmental pressures and governance challenges. Expansion of impervious surfaces reduces infiltration, increases surface runoff, and disrupts hydrological cycles. Moreover, the decline in agricultural and green spaces not only contributes to LST increases but also affects biodiversity, carbon balance, and rural employment.

Environmental consequences are far-reaching: habitat fragmentation, reduced groundwater recharge, and increased flood risk are all linked to these LULC transitions. The findings align with regional studies that report similar trends across the Indo-Gangetic plain, confirming that Varanasi is part of a larger pattern of peri-urban transformation driven by population growth, real estate development, and economic changes.

8.5 Interlinkages and Cross-Variable Patterns

Cross-variable analysis reveals important synergies. The scatterplot of NDVI vs. LST confirms the well-established inverse correlation: as vegetative greenness declines, surface temperatures rise. This is consistent across seasons and spatial scales, reinforcing the critical role of vegetation in thermal regulation.

Additionally, NDVI anomalies correspond strongly with rainfall anomalies. Years with below-average rainfall (e.g., 2009, 2015, 2018) show corresponding dips in NDVI, affirming the moisture dependency of vegetation in this climatic zone. Dual-axis charts plotting NDVI and rainfall anomalies over time effectively visualize this connection, highlighting the compound effects of climate and land cover change.

Spatially, regions showing both vegetation decline and LST increase correspond with zones of built-up expansion. This triad relationship (NDVI↓, LST↑, Built-up↑) suggests a strong human-environment coupling, wherein urbanization alters biophysical processes in measurable and often detrimental ways.

These patterns validate the use of remote sensing as a robust tool for integrated environmental monitoring. They also emphasize the need for interdisciplinary approaches to urban and ecological planning.

8.6 Comparisons with Previous Studies

The observed results are consistent with several regional and national studies. As noted earlier, [Khanam et al. \(2023\)](#)

documented a decline in NDVI in Varanasi during the last two decades, attributed to urban sprawl and reduced agricultural intensity [ijisrt.com]. Similar findings have been reported in Delhi (Kumar et al., 2022), Patna (Sharma & Rao, 2021), and other fast-growing cities in northern India.

However, some discrepancies arise due to differences in temporal coverage, spatial resolution, and data sources. For instance, some studies using MODIS data may show less detail in urban fringes due to its coarser resolution compared to Landsat.

Differences in classification schemes and threshold values for NDVI/SAVI also affect cross-study comparability. Some studies include values below 0.2, which we excluded to avoid noise from bare soil and water bodies. Nevertheless, the overall patterns and interpretations remain consistent: urban growth is linked to vegetation loss, thermal increase, and ecological stress.

8.7 Addressing Uncertainties and Limitations

Several limitations must be acknowledged. First, sensor differences between Landsat 5 TM and Landsat 8 OLI can lead to slight inconsistencies in reflectance values, affecting derived indices. While atmospheric correction and harmonization were applied, residual discrepancies cannot be entirely ruled out.

Second, classification errors in LULC maps may arise from spectral confusion between similar classes, especially in transitional zones. Accuracy assessments were not included in this study due to data constraints, which introduces some degree of classification uncertainty.

Third, rainfall data from CHIRPS, while high-resolution, is satellite-derived and subject to biases in complex terrains or under dense cloud cover. Ground validation with meteorological station data would enhance reliability but was beyond the present scope.

Finally, temporal gaps in satellite data (especially in the 1990s) and seasonal cloud cover can lead to data omission or reduced image count, particularly for monsoon and winter seasons.

Despite these limitations, the integration of multiple datasets and rigorous preprocessing steps lends confidence to the trends and interpretations presented.

8.8 Policy and Planning Implications

The findings of this study carry critical implications for regional policy and planning. The consistent decline in vegetation and rise in land surface temperature highlight the need for urban green space preservation and expansion. Initiatives such as urban forestry, rooftop gardens, and green corridors can help mitigate UHI effects and improve urban liveability.

Given rainfall variability and potential decline in monsoonal intensity, water resource management must prioritize rainwater harvesting, groundwater recharge, and drought-resilient cropping systems. Early warning systems and agricultural advisories based on remote sensing can also support climate-smart farming.

LULC changes call for stricter zoning regulations and environmental impact assessments before land conversion. Restoration of wetlands and peri-urban cropland can serve as ecological buffers and promote sustainable development.

Finally, integration of remote sensing into urban planning—through real-time monitoring of vegetation, temperature, and land cover—can support evidence-based decision-making. Collaborative platforms involving local authorities, academic institutions, and civil society can ensure that data-driven strategies are implemented effectively.

9. Conclusion

This study presents a comprehensive spatiotemporal assessment of vegetation dynamics, surface temperature variability, precipitation trends, and land use/land cover (LULC) transformations in the Varanasi region over two distinct multi-year periods: 1988–1994 and 2013–2024. Utilizing an integrated geospatial approach grounded in satellite-derived indices—NDVI, SAVI, LST, and CHIRPS rainfall—this research provides critical insights into land-atmosphere interactions and anthropogenic pressures shaping the region's environmental trajectory.

9.1 Vegetation Trends: Degradation and Decline

The normalized difference vegetation index (NDVI) and soil-adjusted vegetation index (SAVI), computed with a consistent threshold of 0.23 to exclude sparsely vegetated pixels and background noise, reveal a noticeable decline in vegetation health and density from the earlier to the later study period. During 1988–1994, NDVI and SAVI values were generally higher across seasons, reflecting healthier vegetative cover—particularly during the monsoon and post-monsoon phases. However, the 2013–2024 period exhibits both spatial contraction and spectral weakening of vegetative vigor, particularly around peri-urban zones.

This reduction in vegetative indices aligns with observed urban encroachment and land conversion. While isolated increases in SAVI in some post-monsoon seasons hint at local reforestation or seasonal agricultural revival, the broader trend points towards vegetative stress and degradation. This transformation, as derived from multi-seasonal analysis across 11 years in each period, highlights the shift from predominantly agrarian land uses towards more impervious and urban landscapes, particularly between 2013 and 2021.

9.2 LST Escalation and Urban Influence

Land Surface Temperature (LST) trends show a consistent and significant increase across all seasons in the later period (2013–2024) compared to 1988–1994. The highest LST escalation occurs during the pre-monsoon season, where surface temperatures have risen by several degrees Celsius, especially in densely built-up areas. This increase in surface temperature correlates strongly with areas exhibiting declining NDVI/SAVI, indicating the well-established inverse relationship between vegetation cover and thermal behaviour. Vegetation functions as a thermal buffer, and its reduction has directly contributed to localized warming and intensification of urban heat island (UHI) effects.

The spatial overlap between areas of high LST and areas showing a decrease in NDVI/SAVI further confirms this inverse pattern. This coupling of thermal and vegetative indicators reinforces the argument that surface warming in Varanasi is not merely a consequence of climatic shifts but is deeply tied to land cover transitions and ecosystem alterations driven by anthropogenic activity.

9.3 Rainfall Variability and Vegetative Stress

Analysis of CHIRPS rainfall data over the study periods, particularly during the monsoon months (June–September), reveals a marginal but consistent declining trend in total seasonal precipitation. While interannual variability exists, the overall reduction in rainfall—especially during critical agricultural windows—likely contributed to reduced vegetation productivity, compounding the effects of land cover changes. The relationship between rainfall and vegetation indices is well established, and the observed dip in monsoon rainfall during 2013–2024 likely exacerbated water stress in both natural and cultivated vegetative systems.

Notably, the reduction in NDVI/SAVI during monsoon seasons in the later period correlates with the marginal decline in precipitation. This linkage emphasizes the vulnerability of vegetation cover to not just land use change but also to climatic variability. It further highlights the compounded stress faced by the ecosystem due to simultaneous anthropogenic and meteorological pressures.

9.4 LULC Dynamics: Expansion of Built-Up and Loss of Natural Cover

Land Use Land Cover (LULC) comparison between the earlier and later datasets underscores the scale of anthropogenic transformation. Between 2005–2006 and 2021, the most striking change is the rapid expansion of built-up land from a combined 119.17 sq.km (urban and rural) to 268.01 sq.km. This expansion has primarily come at the expense of cropland, wetland, and vegetative classes. For instance, cropland has declined significantly from 1,095.89 sq.km to 10,811.00 sq.km when considered across datasets, noting that classification schemes differ slightly but still suggest overall losses in productive agricultural land.

Furthermore, natural cover classes such as shrubland, grassland, and tree cover now occupy reduced extents or have been spatially reconfigured. Wetlands, vital for groundwater recharge and biodiversity support, also show stress. The proliferation of impervious surfaces has multiple environmental implications—altered hydrology, reduced evapotranspiration, increased runoff, and decreased carbon sequestration potential.

These LULC transitions mirror and help explain the patterns observed in NDVI/SAVI and LST datasets. Areas of vegetative loss and built-up expansion align with regions of increased surface temperature and decreased spectral vegetation signals, confirming the systemic interrelation between land cover and land surface behaviour.

9.5 Integrated Analysis: Land-Atmosphere Interaction Insights

One of the central contributions of this study is its integrative methodology—linking remote sensing-derived vegetation indices (NDVI/SAVI), land surface temperature (LST), rainfall variability (CHIRPS), and land cover classification over two comparable temporal slices. This cross-thematic, seasonally resolved framework provides a multi-layered understanding of how climatic and anthropogenic factors co-influence the land surface system.

The combined interpretation of these datasets reveals the interconnected nature of land-atmosphere processes: vegetation decline leads to surface heating, which in turn may feedback into microclimatic stress; rainfall variability influences vegetative resilience; and land cover conversion mediates these

dynamics in both spatial and spectral domains. This integrated approach enhances diagnostic capabilities far beyond what single-variable analyses could achieve.

9.6 Contribution to Regional Knowledge and Methodology

To the best of the author's knowledge, this is among the first multi-decadal, multi-index remote sensing studies of the Varanasi region that simultaneously analyzes vegetation health (via NDVI and SAVI), surface thermal dynamics (LST), precipitation trends (CHIRPS), and LULC change in a harmonized framework. By maintaining consistent thresholds (NDVI/SAVI ≥ 0.23), applying robust seasonal classification, and ensuring spatial alignment across datasets, the study ensures comparability and methodological rigor.

The dual-period approach—focusing on 1988–1994 and 2013–2024—avoids the pitfalls of isolated year comparisons and instead enables trend assessments grounded in climatic and developmental phases. The temporal design also accommodates sensor transitions (Landsat 5 TM to Landsat 8 OLI) while acknowledging their inherent differences.

9.7 Acknowledging Uncertainty and Limitations

While the study applies standardized preprocessing and thresholding techniques, several limitations must be acknowledged. First, differences in sensor characteristics between Landsat 5 and Landsat 8 (e.g., radiometric resolution, spectral band adjustments) may introduce slight variations in NDVI and SAVI values despite efforts at standardization. Second, the accuracy of LULC classification is contingent on training data quality, algorithm selection, and seasonal imagery availability—factors that can affect class delineation and area estimation.

Third, the CHIRPS rainfall product, while offering high spatial coverage, is based on interpolated station data and satellite estimation; thus, potential biases or underrepresentation of local-scale anomalies may exist. Lastly, the absence of socio-economic or ground-truth data limits the ability to fully validate and contextualize observed trends.

9.8 Policy Implications and Practical Recommendations

The findings have direct implications for regional planning and resource management. The documented loss of vegetative cover and simultaneous rise in LST underscore the urgent need for policies promoting urban greening, conservation of remaining natural habitats, and the establishment of ecological buffers around expanding urban cores. Regulatory frameworks should aim to limit unchecked sprawl and prioritize the integration of green infrastructure in urban design.

Given rainfall variability and its effect on vegetation productivity, improved water resource management—through rainwater harvesting, wetland restoration, and efficient irrigation practices—should be pursued. Seasonal vegetation and temperature monitoring can also inform early warning systems for droughts and heatwaves, improving adaptive capacities in the face of climate variability.

9.9 Directions for Future Research

This study lays the foundation for further inquiry into land-climate interactions in semi-urban and rural transition zones. Future work can benefit from incorporating additional biophysical indicators such as soil moisture indices (e.g., SMAP data), evapotranspiration estimates, or groundwater depth

records. Socio-economic layers—population growth, migration patterns, land tenure, and development plans—can further enrich the analysis, offering a more comprehensive understanding of the drivers behind environmental change.

Temporal granularity can also be enhanced with higher-resolution datasets (e.g., Sentinel series or commercial imagery), enabling finer-scale assessments of microclimatic variation and urban morphology. Machine learning-based classification and change detection methods may also offer improved accuracy in LULC mapping.

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